

Normalized band-limited translates show that $\mathcal{D}_E$ is never Montel

Solution packet for arXiv:1310.4047

22 July 2026

Status: claimed full solution; likely valid. For every translation-invariant Banach space of tempered distributions E in the sense of Dimovski–Pilipović–Vindas, the Fréchet space $\mathcal{D}_E$ is not Montel. No solidity, reflexivity, or boundedness of the translation weight is assumed.

1 Source question

Dimovski, Pilipović, and Vindas introduce a translation-invariant Banach space E of tempered distributions and the associated test-function space $\mathcal{D}_E$ in [1]. In Remark 6 on page 17 they ask:

“In general: Can $\mathcal{D}_E$ be Montel?”

The source passage is reproduced below at full readable width.

Remark 6. Schwartz has pointed out [28, p. 200] that the spaces $\mathcal{D}_{L^p}$ are not Montel. Remark 5 then yields that $\mathcal{D}_{L^p_\omega}$ are not Montel either, $1 \leq p < \infty$. The spaces $\mathcal{B}_\omega$ and $\mathcal{B}'_\omega$ can never be Montel because they are not reflexive. When ω is bounded, it is easy to see that $\mathcal{D}_E$ is never Montel. In fact, if $\varphi \in \mathcal{D}(\mathbb{R}^n)$ is such that $\varphi(x) = 0$ for $|x| \geq 1/2$ and $\theta \in \mathbb{R}^n$ is a unit vector, then $\{T_{-j\theta}\varphi\}_{j=0}^\infty$ is a bounded sequence in $\mathcal{D}_E$ without any accumulation point, as follows from the continuous inclusion $\mathcal{D}_E \rightarrow \mathcal{B}$. In general: Can $\mathcal{D}_E$ be Montel?

Remark 7. When E is reflexive, the space $\mathcal{D}'_E$ is barreled, as follows from Proposition 9 because a reflexive space is barreled. In the general case: Is the space $\mathcal{D}'_{E'_*}$ barreled?

Debrouwere later restated the question as Problem 5.8 and proved that $\mathcal{D}_E$ is not Montel when E is a *solid* translation-invariant Banach space of distributions [2, Problem 5.8 and Theorem 5.9]. That paper explicitly says that the general case was not known. The proof below removes solidity.

2 Setting and definitions

We use the source convention $T_x f = f(\cdot + x)$. A translation-invariant Banach space of tempered distributions is a Banach space E for which

$$\mathcal{S}(\mathbb{R}^n) \hookrightarrow E \hookrightarrow \mathcal{S}'(\mathbb{R}^n), \quad T_x E \subset E,$$

the translation orbit maps are continuous, and for some $M > 0$ and $\tau \geq 0$,

$$\|T_x f\|_E \leq M(1 + |x|)^\tau \|f\|_E.$$

Its translation weight is

$$\omega(x) = \|T_{-x}\|_{\mathcal{L}(E)} \leq M(1 + |x|)^\tau.$$

The source proves that E is a Banach module over the Beurling algebra L_ω^1 and that

$$\|u * f\|_E \leq \|u\|_{1,\omega} \|f\|_E. \quad (1)$$

The Fréchet space in the question is

$$\mathcal{D}_E = \{f \in \mathcal{S}'(\mathbb{R}^n) : \partial^\alpha f \in E \text{ for every } \alpha \in \mathbb{N}_0^n\},$$

with increasing seminorms

$$p_N(f) = \max_{|\alpha| \leq N} \|\partial^\alpha f\|_E.$$

The embeddings $\mathcal{D}_E \hookrightarrow E \hookrightarrow \mathcal{S}'(\mathbb{R}^n)$ are continuous [1, Proposition 7].

3 Proof intuition

Unnormalized translates need not form a bounded set because the translation weight may grow. Normalize a translate $T_x \phi$ by its own E -norm. This fixes its zeroth seminorm at one, but could in principle make higher derivatives unbounded. A band-limited Schwartz function prevents that: every derivative of ϕ is a fixed L_ω^1 convolution of ϕ . The module estimate then controls every derivative of every normalized translate uniformly.

The normalization also cannot grow faster than a polynomial, because applying T_{-x} recovers ϕ . Against a Schwartz test function, however, the translate $T_x \phi$ decays faster than every polynomial. Thus the normalized translates converge to zero in $\mathcal{S}'(\mathbb{R}^n)$, although their E -norms remain one. They therefore have no convergent subsequence in $\mathcal{D}_E$.

4 Band-limited derivative control

Fix the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} e^{-ix \cdot \xi} f(x) dx.$$

Lemma 1. *There is a nonzero $\phi \in \mathcal{S}(\mathbb{R}^n)$ such that, for every multi-index α , there is $u_\alpha \in \mathcal{S}(\mathbb{R}^n) \subset L_\omega^1$ satisfying*

$$\partial^\alpha \phi = u_\alpha * \phi.$$

Consequently, the family

$$g_x := \frac{T_x \phi}{\|T_x \phi\|_E}, \quad x \in \mathbb{R}^n,$$

is bounded in $\mathcal{D}_E$.

Proof. Choose a nonzero $\widehat{\phi} \in C_c^\infty(\mathbb{R}^n)$ and let $\phi = \mathcal{F}^{-1} \widehat{\phi}$. Choose $\chi \in C_c^\infty(\mathbb{R}^n)$ with $\chi = 1$ on $\text{supp } \widehat{\phi}$, and define

$$\widehat{u}_\alpha(\xi) = \chi(\xi)(i\xi)^\alpha.$$

Then $u_\alpha \in \mathcal{S}(\mathbb{R}^n)$. Since ω is polynomially bounded, $u_\alpha \in L_\omega^1$. Fourier transformation gives

$$\widehat{u_\alpha * \phi} = \chi(\xi)(i\xi)^\alpha \widehat{\phi}(\xi) = (i\xi)^\alpha \widehat{\phi}(\xi) = \widehat{\partial^\alpha \phi}.$$

Translations commute with derivatives and convolution, so (1) implies

$$\|\partial^\alpha g_x\|_E = \frac{\|u_\alpha * T_x \phi\|_E}{\|T_x \phi\|_E} \leq \|u_\alpha\|_{1,\omega}.$$

For every fixed N , taking the maximum over $|\alpha| \leq N$ proves that $\sup_x p_N(g_x) < \infty$. The denominator is nonzero because translations are invertible and $\phi \neq 0$. $\square$

5 Full negative answer

Theorem 1. *Let E be any translation-invariant Banach space of tempered distributions in the sense of [1]. Then $\mathcal{D}_E$ is not Montel.*

Proof. Let ϕ and g_x be as in Lemma 1. Since $T_{-x}T_x\phi = \phi$ and $\omega(x) = \|T_{-x}\|_{\mathcal{L}(E)}$,

$$\|\phi\|_E \leq \omega(x)\|T_x\phi\|_E, \quad \frac{1}{\|T_x\phi\|_E} \leq \frac{M(1+|x|)^\tau}{\|\phi\|_E}. \quad (2)$$

For every $\psi \in \mathcal{S}(\mathbb{R}^n)$, the function

$$x \mapsto \langle T_x\phi, \psi \rangle$$

is a Schwartz function of x : after a reflection it is the convolution of two Schwartz functions. Combining this rapid decay with (2) yields

$$\langle g_x, \psi \rangle = \frac{\langle T_x\phi, \psi \rangle}{\|T_x\phi\|_E} \longrightarrow 0 \quad (|x| \rightarrow \infty).$$

Thus $g_x \rightarrow 0$ in the weak topology $\sigma(\mathcal{S}'(\mathbb{R}^n), \mathcal{S}(\mathbb{R}^n))$.

Choose any sequence x_j with $|x_j| \rightarrow \infty$. By Lemma 1, (g_{x_j}) is bounded in $\mathcal{D}_E$. If $\mathcal{D}_E$ were Montel, the closure of this bounded set would be compact. Because $\mathcal{D}_E$ is Fréchet and hence metrizable, (g_{x_j}) would have a subsequence converging in $\mathcal{D}_E$. The continuous embedding $\mathcal{D}_E \hookrightarrow \mathcal{S}'(\mathbb{R}^n)$ and the weak convergence above force its limit to be zero. But

$$p_0(g_{x_j}) = \|g_{x_j}\|_E = 1$$

for every j , so no subsequence can converge to zero in $\mathcal{D}_E$. This contradiction proves the theorem. $\square$

6 Verification and scope

The proof uses only four source-paper facts: polynomial growth of ω , the convolution estimate (1), the definition of the $\mathcal{D}_E$ seminorms, and the continuous embedding $\mathcal{D}_E \hookrightarrow \mathcal{S}'(\mathbb{R}^n)$. The normalization estimate (2) was checked against the source sign convention $\omega(x) = \|T_{-x}\|$. No numerical or computer-assisted step enters the proof.

A bounded novelty search performed on 22 July 2026 covered arXiv and web searches for arXiv ids 1310.4047 and 1901.10041, the exact source question, “ $\mathcal{D}_E$ is never Montel,” “general TIBD,” the paper titles, and the relevant authors. It found the 2019 solid-space theorem and its explicit statement that the general case remained open, but no later general solution. This is evidence, not an exhaustive bibliographic guarantee; novelty confidence is therefore *moderate to high*.

The theorem fully answers Remark 6 negatively. It does not address the separate barreledness question in Remark 7. Recommended human review: **verify and circulate**, with particular attention to the multiplier identity in Lemma 1 and the compactness-to-subsequence step.

References

- [1] P. Dimovski, S. Pilipović, and J. Vindas, *New distribution spaces associated to translation-invariant Banach spaces*, Monatsh. Math. **177** (2015), 495–515; arXiv:1310.4047.
- [2] A. Debrouwere, *Quasinormable C_0 -groups and translation-invariant Fréchet spaces of type $\mathcal{D}_E$* , Results Math. **74** (2019), Article 135; arXiv:1901.10041.